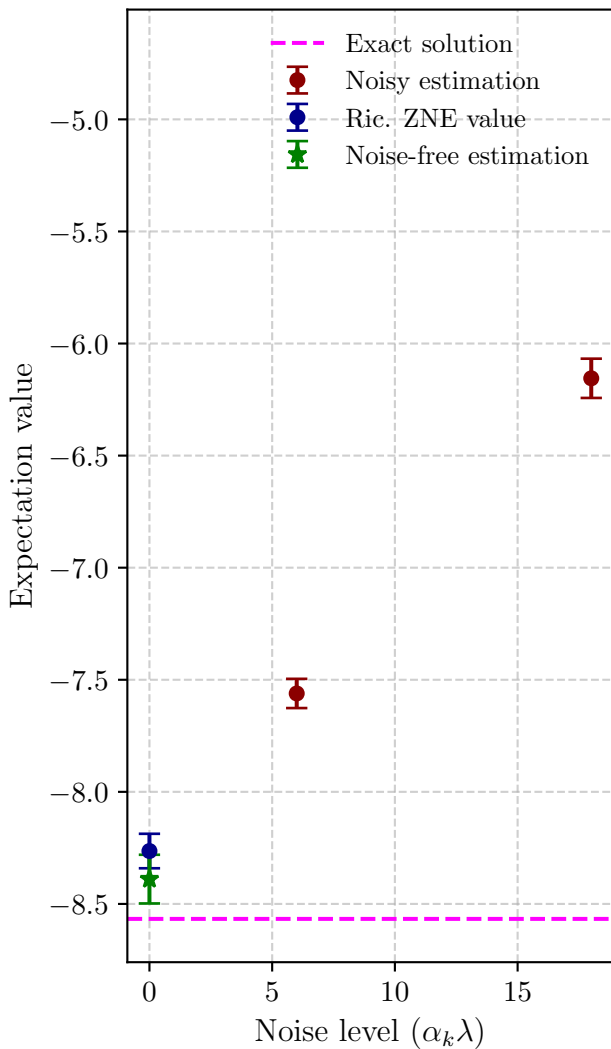
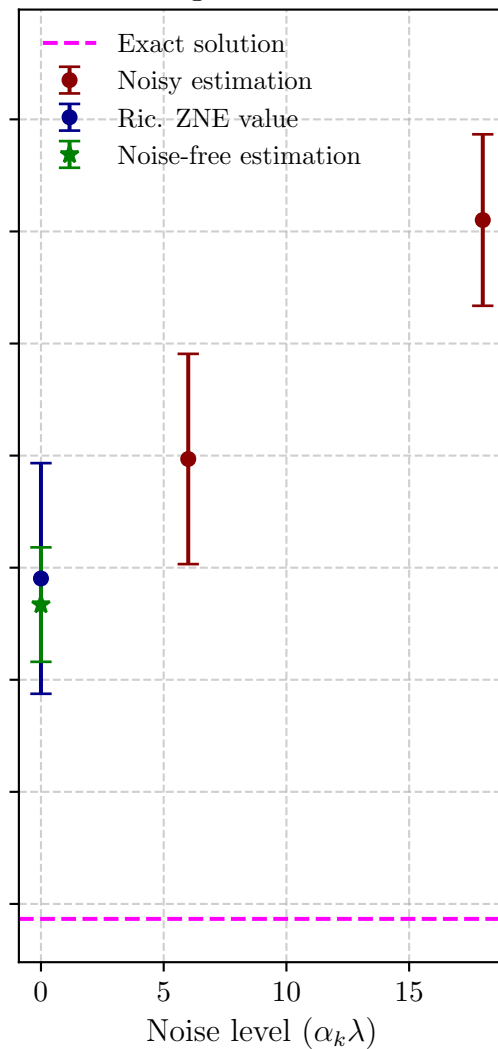


| Description                     | Simulation           | Mean $\pm$ Std         |
|---------------------------------|----------------------|------------------------|
| xy-ric2                         | vqe                  | -7.56107 $\pm$ 0.06520 |
| xy-ric2                         | redundant (noise=6)  | -7.56107 $\pm$ 0.06520 |
| xy-ric2                         | redundant (noise=18) | -6.15518 $\pm$ 0.08758 |
| xy-ric2                         | zne                  | -8.26402 $\pm$ 0.07690 |
| xy-ric2                         | noiseoff             | -8.38929 $\pm$ 0.10862 |
|                                 |                      |                        |
| ising-ric2                      | vqe                  | -6.51507 $\pm$ 0.46899 |
| ising-ric2                      | redundant (noise=6)  | -6.51507 $\pm$ 0.46899 |
| ising-ric2                      | redundant (noise=18) | -5.44914 $\pm$ 0.38250 |
| ising-ric2                      | zne                  | -7.04804 $\pm$ 0.51449 |
| ising-ric2                      | noiseoff             | -7.16465 $\pm$ 0.25530 |
|                                 |                      |                        |
| heisenberg-ric2                 | vqe                  | -6.26219 $\pm$ 0.17777 |
| heisenberg-ric2                 | redundant (noise=6)  | -6.26219 $\pm$ 0.17777 |
| heisenberg-ric2                 | redundant (noise=18) | -4.92419 $\pm$ 0.22076 |
| heisenberg-ric2                 | zne                  | -6.93119 $\pm$ 0.16633 |
| heisenberg-ric2                 | noiseoff             | -6.72634 $\pm$ 0.32239 |
|                                 |                      |                        |
| xy-ric3-uniform-scaling         | vqe                  | -7.56107 $\pm$ 0.06520 |
| xy-ric3-uniform-scaling         | redundant (noise=6)  | -7.56107 $\pm$ 0.06520 |
| xy-ric3-uniform-scaling         | redundant (noise=18) | -6.15518 $\pm$ 0.08758 |
| xy-ric3-uniform-scaling         | redundant (noise=30) | -5.04735 $\pm$ 0.12295 |
| xy-ric3-uniform-scaling         | zne                  | -8.37580 $\pm$ 0.08182 |
| xy-ric3-uniform-scaling         | noiseoff             | -8.38929 $\pm$ 0.10862 |
|                                 |                      |                        |
| ising-ric3-uniform-scaling      | vqe                  | -6.51507 $\pm$ 0.46899 |
| ising-ric3-uniform-scaling      | redundant (noise=6)  | -6.51507 $\pm$ 0.46899 |
| ising-ric3-uniform-scaling      | redundant (noise=18) | -5.44914 $\pm$ 0.38250 |
| ising-ric3-uniform-scaling      | redundant (noise=30) | -4.56370 $\pm$ 0.32425 |
| ising-ric3-uniform-scaling      | zne                  | -7.11572 $\pm$ 0.52373 |
| ising-ric3-uniform-scaling      | noiseoff             | -7.16465 $\pm$ 0.25530 |
|                                 |                      |                        |
| heisenberg-ric3-uniform-scaling | vqe                  | -6.26219 $\pm$ 0.17777 |
| heisenberg-ric3-uniform-scaling | redundant (noise=6)  | -6.26219 $\pm$ 0.17777 |
| heisenberg-ric3-uniform-scaling | redundant (noise=18) | -4.92419 $\pm$ 0.22076 |
| heisenberg-ric3-uniform-scaling | redundant (noise=30) | -3.86627 $\pm$ 0.25204 |
| heisenberg-ric3-uniform-scaling | zne                  | -7.03622 $\pm$ 0.16244 |
| heisenberg-ric3-uniform-scaling | noiseoff             | -6.72634 $\pm$ 0.32239 |
|                                 |                      |                        |
| xy-ric4-uniform-scaling         | vqe                  | -7.56107 $\pm$ 0.06520 |
| xy-ric4-uniform-scaling         | redundant (noise=6)  | -7.56107 $\pm$ 0.06520 |
| xy-ric4-uniform-scaling         | redundant (noise=18) | -6.15518 $\pm$ 0.08758 |
| xy-ric4-uniform-scaling         | redundant (noise=30) | -5.04735 $\pm$ 0.12295 |
| xy-ric4-uniform-scaling         | redundant (noise=42) | -4.16757 $\pm$ 0.15007 |
| xy-ric4-uniform-scaling         | zne                  | -8.39767 $\pm$ 0.08306 |
| xy-ric4-uniform-scaling         | noiseoff             | -8.38929 $\pm$ 0.10862 |
|                                 |                      |                        |
| ising-ric4-uniform-scaling      | vqe                  | -6.51507 $\pm$ 0.46899 |
| ising-ric4-uniform-scaling      | redundant (noise=6)  | -6.51507 $\pm$ 0.46899 |
| ising-ric4-uniform-scaling      | redundant (noise=18) | -5.44914 $\pm$ 0.38250 |
| ising-ric4-uniform-scaling      | redundant (noise=30) | -4.56370 $\pm$ 0.32425 |
| ising-ric4-uniform-scaling      | redundant (noise=42) | -3.82308 $\pm$ 0.28550 |
| ising-ric4-uniform-scaling      | zne                  | -7.12687 $\pm$ 0.52568 |
| ising-ric4-uniform-scaling      | noiseoff             | -7.16465 $\pm$ 0.25530 |
|                                 |                      |                        |
| heisenberg-ric4-uniform-scaling | vqe                  | -6.26219 $\pm$ 0.17777 |
| heisenberg-ric4-uniform-scaling | redundant (noise=6)  | -6.26219 $\pm$ 0.17777 |
| heisenberg-ric4-uniform-scaling | redundant (noise=18) | -4.92419 $\pm$ 0.22076 |
| heisenberg-ric4-uniform-scaling | redundant (noise=30) | -3.86627 $\pm$ 0.25204 |
| heisenberg-ric4-uniform-scaling | redundant (noise=42) | -3.02209 $\pm$ 0.26938 |
| heisenberg-ric4-uniform-scaling | zne                  | -7.05696 $\pm$ 0.16129 |
| heisenberg-ric4-uniform-scaling | noiseoff             | -6.72634 $\pm$ 0.32239 |
|                                 |                      |                        |

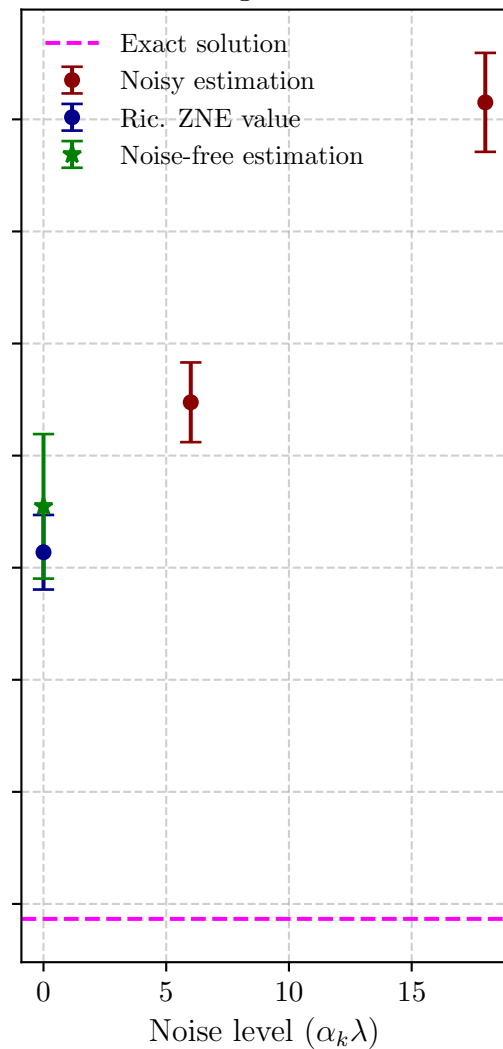
XY ansatz ric-2



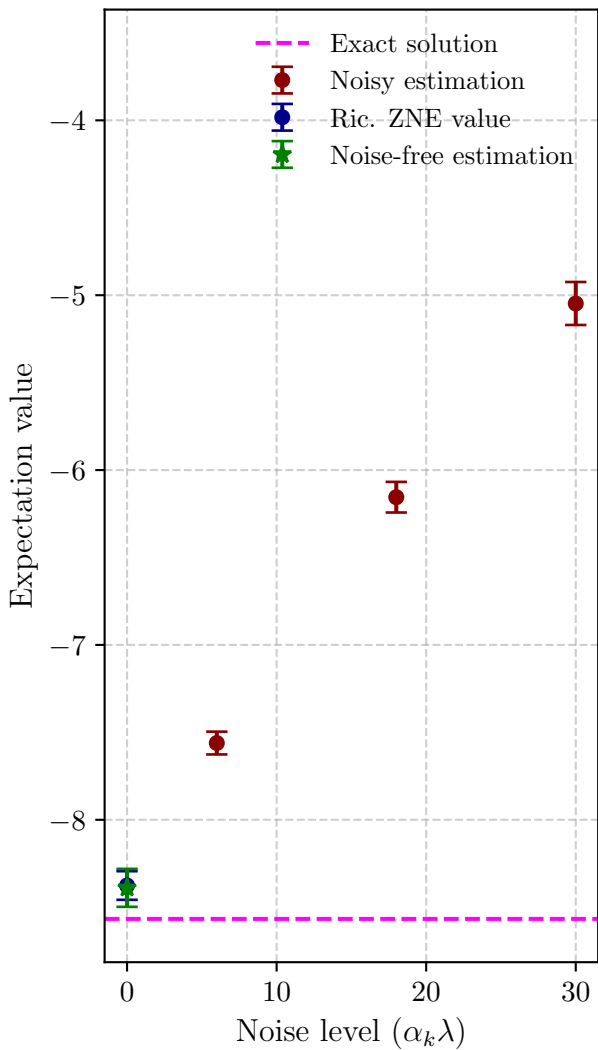
Ising ansatz ric-2



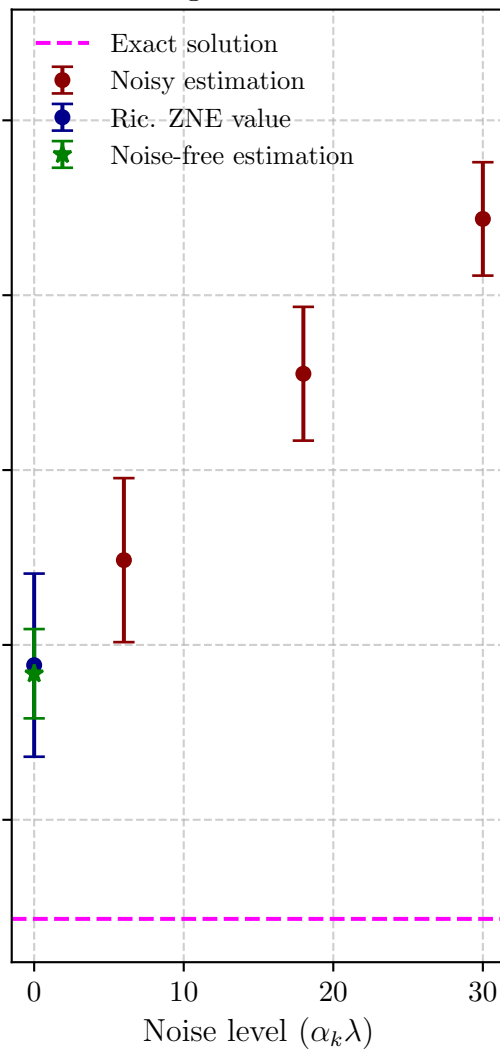
Heisenberg ansatz ric-2



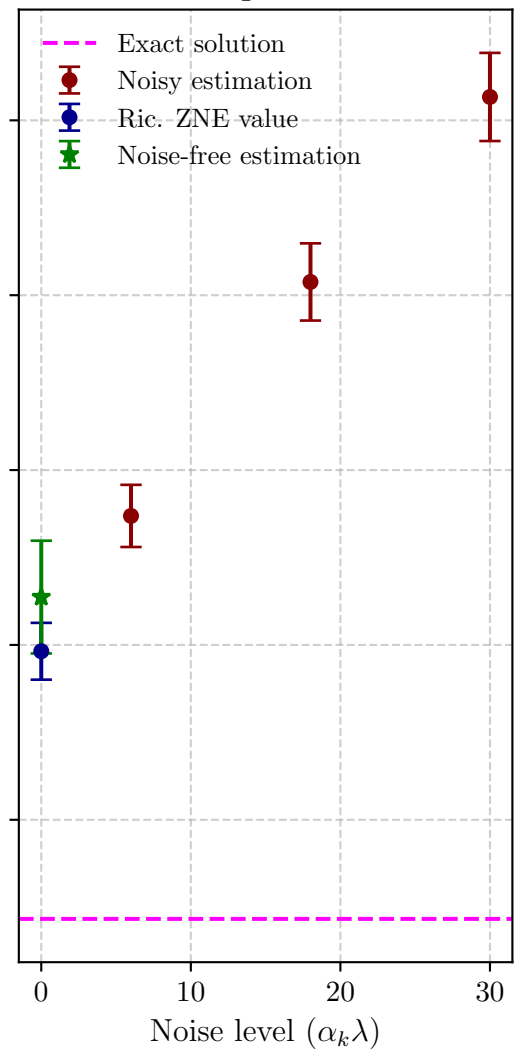
XY ansatz ric-3



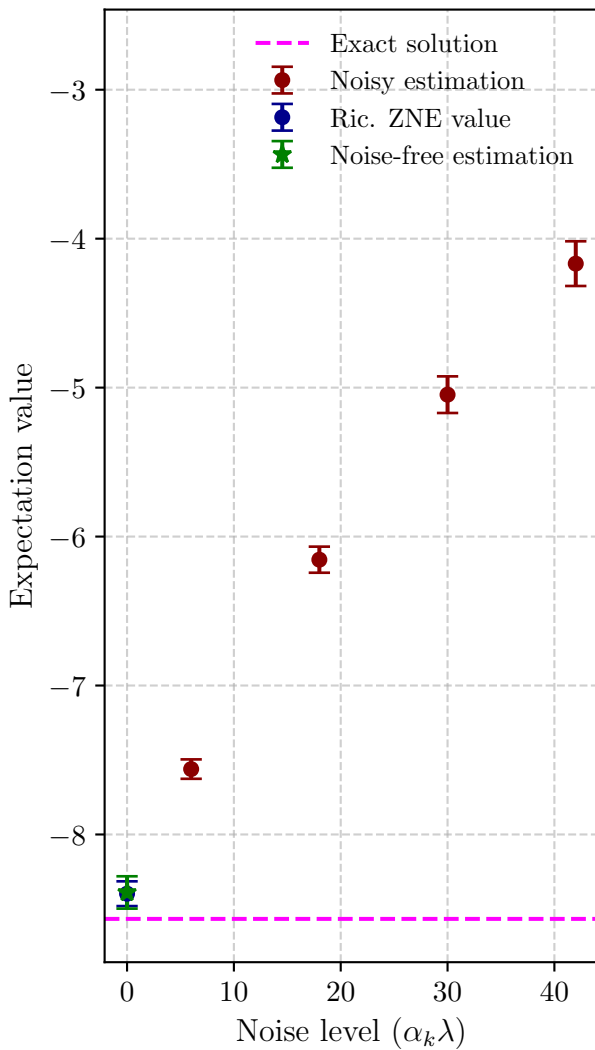
Ising ansatz ric-3



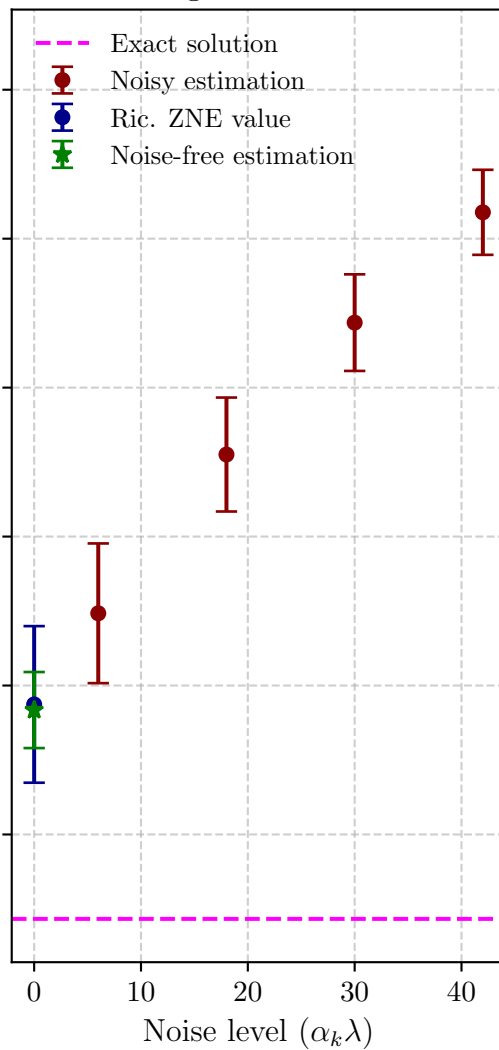
Heisenberg ansatz ric-3



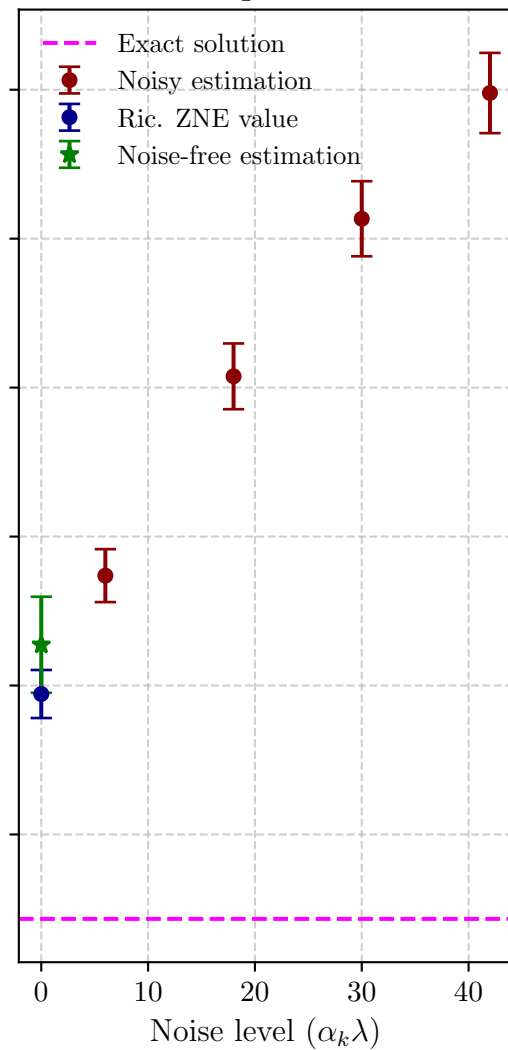
XY ansatz ric-4



Ising ansatz ric-4



Heisenberg ansatz ric-4



# XY VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | xy_noisefree_time_evo               |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | xy-iss                              |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | dephasing                           |
| noise_profile.noise_prob                 | [0.001, 0.001, 0.0, 0.0]            |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 1, 1]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |

ISING VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | ising_noisefree_time_evo            |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | ising                               |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | dephasing                           |
| noise_profile.noise_prob                 | [0.001, 0.001, 0.0, 0.0]            |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 1, 1]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |

# HEISENBERG VQE CONFIGURATION

| Parameter                                | Value                               |
|------------------------------------------|-------------------------------------|
| run                                      | vqe                                 |
| nqubits                                  | 7                                   |
| state                                    | dmatrix                             |
| output.file_name_prefix                  | heisenberg_noise_free_time_evo      |
| output.draw.status                       | True                                |
| output.draw.fig_dpi                      | 100                                 |
| output.draw.type                         | png                                 |
| observable.def                           | ising                               |
| observable.coefficients.cn               | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| observable.coefficients.bn               | [1.0, 1.0, 1.0, 1.0, 1.0, 1.0, 1.0] |
| observable.coefficients.r                | 1                                   |
| ansatz.layer                             | 30                                  |
| ansatz.gateset                           | 1                                   |
| ansatz.ugate.type                        | heisenberg                          |
| ansatz.ugate.coefficients.cn             | [0.5, 0.5, 0.5, 0.5, 0.5, 0.5]      |
| ansatz.ugate.coefficients.bn             | [0, 0, 0, 0, 0, 0, 0]               |
| ansatz.ugate.coefficients.r              | 0                                   |
| ansatz.ugate.time.min                    | 0.0                                 |
| ansatz.ugate.time.max                    | 10.0                                |
| vqe.iteration                            | 10                                  |
| vqe.optimization.status                  | True                                |
| vqe.optimization.algorithm               | SLSQP                               |
| vqe.optimization.constraint              | True                                |
| init_param.value                         | random                              |
| noise_profile.status                     | True                                |
| noise_profile.type                       | dephasing                           |
| noise_profile.noise_prob                 | [0.001, 0.001, 0.0, 0.0]            |
| noise_profile.noise_on_init_param.status | False                               |
| noise_profile.noise_on_init_param.value  | 0                                   |
| redundant.identity_factors               | [[0, 0, 0, 0], [1, 1, 1, 1]]        |
| zne.method                               | richardson                          |
| zne.degree                               | 1                                   |
| zne.sampling                             | default                             |
| zne.data_points                          |                                     |